Arista Access Node KR-ACC-20 Optical Transport Failure SOP

Node: KR-ACC-20 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_3940

Description

Optical transport failures on the Arista Access Node KR-ACC-20 in the KR Puram Access site can occur due to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. This is primarily attributed to the inherent limitations of fiber optic communication systems, where signal attenuation over long distances necessitates the use of Optical Transport Network (OTN) equipment. In the context of the Arista Access Node, the OTUk (Optical Transport Unit k) frames are transmitted over the fiber optic cable, and any degradation or failure in the OTUk frames can trigger the ALARM_3940 alarm code. The ALARM_3940 is a Major alarm code indicating a significant fault in the optical transport path, which can impact the overall network performance and reliability.

Pre-requisite Checks

- pre_requisite: Verify that the KR Puram Access site is physically accessible and the Access Node KR-ACC-20 is powered on., description: Ensure that the site is accessible to perform the necessary diagnostic and resolution steps.

- pre_requisite: Check the redundant link status to verify if the failure is isolated to the primary link or if the redundant link is also affected., description: Verify the configuration and health of the redundant link to determine the scope of the failure.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

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Command: show interface eth1/1 transceiver, Expected Output: Optical Transceiver Information, Description: This command verifies the optical transceiver information for the eth1/1 interface, including the SFP/XFP type, temperature, and voltage information. The output will help identify any issues with the optical transceiver that may be contributing to the ALARM_3940 alarm code.

Command: show interface eth1/1 txpower, Expected Output: Transmit Power Information, Description: This command verifies the transmit power for the eth1/1 interface, including the current transmit power level and the maximum allowed transmit power level. The output will help identify any issues with the transmit power that may be
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contributing to the ALARM_3940 alarm code.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace interface eth1/1 transceiver, Description: Replace the SFP/XFP module with a new one to resolve the ALARM_3940 alarm code. Verify that the new module is properly installed and configured before proceeding.

Step 2: Step: Verify the fiber optic cable, Command: show interface eth1/1 cable, Description: Verify the fiber optic cable connection to the Access Node KR-ACC-20. Check for any signs of physical damage or wear on the fiber optic cable that may be contributing to the ALARM_3940 alarm code.

Post-Check Verification

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Command: show interface eth1/1 status, Expected Output: Interface Status Information, Description: This command verifies the interface status information for the eth1/1 interface, including the current link status and any error counters. The output will help verify that the ALARM 3940 alarm code has been cleared.

Command: show alarm history, Expected Output: Alarm History Information, Description: This command verifies the alarm history information for the Access Node KR-ACC-20, including the current alarm status and any previous alarm events. The output will help verify that the ALARM_3940 alarm code has been cleared.
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Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: replace interface eth1/1 transceiver, Description: Restore the original SFP/XFP module to its original configuration to rollback any changes made during the resolution step.

- Step: Verify the fiber optic cable, Command: show interface eth1/1 cable, Description: Verify the fiber optic cable connection to the Access Node KR-ACC-20 to ensure that it is properly connected and configured.